

Ruthvek Kannan

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Summary

Computer Science undergraduate specializing in applied AI/ML engineering — multi-agent orchestration, retrieval-augmented generation, and full-stack deployment — with sandboxed execution, observability, and cost governance built into every system I ship.

Education

Symbiosis Institute of Technology, Nagpur (Symbiosis International Deemed University)

Bachelor of Technology in Computer Science & Engineering

2022 – 2026

Nagpur, India

Experience

Apoliums Infotech India Pvt. Ltd.

Dec 2025 – Jun 2026

Full-Stack Web & Mobile Application Development Intern

India

- Developed and shipped full-stack web and mobile applications across multiple client domains using JavaScript/TypeScript and mobile frameworks.
- Integrated LLM and AI capabilities into production business applications, connecting REST APIs to third-party AI providers.
- Collaborated with cross-functional engineering teams to design responsive UIs, integrate cloud services, and deploy scalable backend solutions.

Projects

AEgis – Multi-Agent Software Engineering Framework

[Live Demo](#) | [GitHub](#)

Python, FastAPI, LangGraph, React, Docker, PostgreSQL, pgvector

- Architected a LangGraph-orchestrated multi-agent DAG (Supervisor, Planner, Coder, Critic, Synthesizer) enforcing a Red-Green-Refactor verification loop that requires proof of a failing test before any fix is accepted.
- Built isolated, non-root Docker sandboxing for all LLM-generated code execution, with a graceful local-subprocess fallback ensuring uptime independent of container infrastructure.
- Engineered real-time cost governance via OpenTelemetry with hard-dollar budget caps and API key load-balancing; deployed the full stack (FastAPI, React/Vite/SSE) on free-tier infrastructure (Vercel, Render, Supabase).

Athenis AI – Enterprise Hybrid-RAG Platform

[Live Demo](#) | [GitHub](#)

Python, FastAPI, Next.js, Celery, PostgreSQL, Redis, Kubernetes

- Built a hybrid-search RAG engine combining pgvector cosine similarity with PostgreSQL full-text search (BM25), merged via Reciprocal Rank Fusion for high-precision retrieval.
- Designed a provider-agnostic LLM abstraction layer (LiteLLM) supporting Gemini, OpenAI, Anthropic, and local Ollama models with no code changes required to switch providers.
- Implemented JWT-based RBAC with Admin/User segregation and multi-tenant billing that tracks per-workspace token usage against hard and soft budget quotas.

Proactive Conversational AI Assistant for Data Science Workflows

Academic Capstone

Python, Gemini API, Whisper, PyCaret, Streamlit

- Co-architected an event-driven assistant and designed **Project Watchdog**, an autonomous file-monitoring service that detects new datasets in real time and proactively triggers analysis via LLM-based intent recognition and voice interaction.
- Automated exploratory data analysis: LLM-generated visualization code and AI-written interpretation are executed and compiled into a multi-page PDF report via fpdf2.
- Measured a ~90% reduction in time-to-insight (15–20 min manual workflow vs. 1.5–2 min automated), validated in a comparative user study.

Technical Skills

Languages: Python, TypeScript, JavaScript (ES6+), Dart, C, SQL, HTML5/CSS3

AI/ML & LLM Engineering: LangGraph, LangChain, LiteLLM, RAG (Hybrid Search, RRF), Multi-Agent Orchestration, AutoML (PyCaret), NLP/Intent Recognition, Prompt Engineering

Backend: FastAPI, REST APIs, Server-Sent Events, Celery, SQLAlchemy

Frontend & Creative Engineering: React, Next.js, Three.js/React Three Fiber, Framer Motion, GSAP, Tailwind CSS, Streamlit, RSC/SSG, Web Performance & SEO, Accessibility (a11y)

Data & Infrastructure: PostgreSQL, pgvector, Redis, Docker, Kubernetes, OpenTelemetry, Prometheus, Grafana

Cloud & Tools: AWS, Firebase, Vercel, Render, Supabase, Git, GitHub Actions (CI/CD), LaTeX, Power BI

Research & Certifications

Research: “Architectural Innovations for AI Chips, Testing, and High Performance Computing” – published research paper.

Certifications: AWS Cloud Foundations, AWS Cloud Security, Oracle Database Foundations, Oracle Java Fundamentals, Red Hat System Administration I, ServiceNow CSA, ServiceNow CAD, Infosys Big Data.